Response to Reviewer 1

We thank the reviewer for a careful and constructive reading of the manuscript. The comments on novelty framing, the oracle-efficiency metric, the fairness of baseline comparisons, the strength of the causal claims, and the clarity of the figures have substantially improved the paper. The manuscript has been revised throughout; all changes are summarised below, with pointers to the relevant sections, tables, and figures of the revised manuscript. Page numbers refer to the revised compiled PDF.

Summary of the main changes

- The contribution is now framed explicitly as an intervention-selection layer on top of an existing attribution backend, not a new attribution method (Introduction, p. 2; Section "Comparison with Published Circuit Discovery Methods", Table 11, p. 31).
- The $> 100\%$ "oracle efficiency" has been replaced by two clearly separated metrics: a bounded oracle efficiency ($\leq 100\%$) for ablation-only selectors, and a Relative Cumulative KL (RCK) for steering-enabled selectors (Section "Metrics", p. 16–17).
- A dose–response steering sweep and a random-feature control were added, and the causal claims were weakened accordingly (Section "Feature Steering", p. 13; Results, p. 24).
- Figures 1 and 2 were redrawn as clean raster diagrams; the remaining figures were given shared, non-overlapping legends.
- The oracle rows of the KL tables now report the actual oracle KL values.
- Action-matched (steering-enabled) variants of random, greedy, and bandit were added, plus a random-action control, and the Llama multi-step shortfall is now treated as a limitation.

Point-by-point responses

> ***Comment 1.*** *The novelty of the proposed framework should be clarified. It is not fully clear whether the main contribution is a new circuit discovery method or an active intervention-selection strategy built on existing attribution graph tools.*

Response: The contribution is the latter, and the manuscript now states this plainly. The revised Introduction (p. 2) describes the work as "an intervention-selection layer that sits on top of an existing attribution backend", and the comparison section (Table 11, p. 31) explicitly lists `circuit-tracer` as the attribution backend and positions the agent as the selection policy operating over its candidate set. The novelty claimed is the active-inference POMDP formulation of which feature and which intervention type to test next under a fixed budget, not the construction of the attribution graph itself.

> ***Comment 2.*** *The "oracle efficiency" metric needs a clearer definition. Reporting values above 100%, such as 1255%, is confusing if the oracle is an upper bound. This should be mathematically explained or renamed.*

Response: The metric has been split and renamed. For ablation-only selectors, every method and the oracle draw from the same quantity (ablation KL), so the ratio is a genuine bounded efficiency $\eta \in [0, 100]\%$ (Equation for η, Section "Metrics", p. 16). For the full multi-action agent, which may steer, the cumulative KL is compared with the ablation oracle through a separate quantity, the Relative Cumulative KL (RCK, Equation for RCK, p. 17), explicitly labelled as a relative amplification factor and never as an efficiency. The 1255% figure now appears only as an RCK value in Table 3 (p. 22) and is described as steering-driven amplification rather than super-oracle discovery.

> ***Comment 3.*** *The causal claims should be stated more cautiously. Feature steering with 10× activation scaling may cause out-of-distribution effects, so additional controls such as smaller scaling factors and random-feature comparisons would strengthen the claim.*

Response: Both controls were added. The steering protocol now sweeps $m \in \{0, 1.5, 2, 3, 5, 10\}$ and compares each circuit-selected feature against a matched random-feature control steered at the same multipliers (Section "Feature Steering", p. 13). The Results section (p. 24) reports that the selectivity advantage of circuit features over the control is not statistically significant at large multipliers (one-sided Fisher exact test), so the steering result is now framed as a controllability demonstration rather than proof of privileged selectivity. A complementary top-k token analysis shows that large multipliers amplify coherent concept tokens rather than only degrading the model, which is reported alongside the dose–response curve.

> ***Comment 4.*** *Figure 1 should be visually revised. Some arrows/lines appear to cross through each other, which makes the workflow difficult to follow.*

Response: Figures 1 and 2 (system architecture and the EFE pipeline) were re-authored as clean raster diagrams (`fig1.png`, `fig2.png`) with non-crossing routing and clear spacing, replacing the previous TikZ figures (Methodology, p. 8 and p. 9).

> ***Comment 5.*** *In Tables 2 and 4, the oracle row reports only 100% efficiency but not the corresponding mean or cumulative KL value. The authors should report the actual oracle KL scores to make the normalization transparent.*

Response: The oracle KL values are now reported. Table 2 (IOI, p. 19) lists the oracle's per-prompt mean KL (0.000862 for Gemma, 0.009755 for Llama), and the surrounding text gives the cumulative-KL denominators (0.0172 and 0.1951). The multi-step table (Table 5, p. 26) likewise reports the oracle cumulative KL (0.0101 Gemma, 0.2447 Llama). The normalisation underlying every efficiency value is therefore explicit.

> ***Comment 6.*** *The POMDP agent performs worse than the bandit in the Llama multi-step test (33.6% vs. 76.4%). This negative result is significant and should be treated as a limitation, not an architectural explanation.*

Response: The Llama multi-step shortfall is now reported as a genuine limitation. Under the fair ablation-only protocol the multi-step results appear in Table 5 (p. 26), and the Discussion ("Cross-Model Generality" and "Limitations", p. 31) states that the agent underperforms on Llama multi-step and that a finer-layer-bin variant was tried but did not recover the gap (Table 5). The result is presented as an open limitation tied to the coarse layer-role discretisation rather than as a feature of the architecture.

> ***Comment 7.*** *Only an ablation version of the POMDP is included; action-matched variants are missing for random, greedy, and bandit. Does the POMDP's success stem from EFE-guided selection or from the steering action producing higher KL? Action-matched versions of the baselines would make the comparison more balanced.*

Response: Action-matched baselines were added. Greedy+steer, Bandit+steer, and Random+steer use the identical multi-action space as the full agent, and a Random-action control randomises the intervention type (Section "Baselines", p. 15). Reported on the common RCK scale (Table 3, p. 22), these show that on Gemma every steering-enabled strategy, including random+steer (888%), produces RCK well above 100%, so the amplification is attributable to the steering action itself rather than to which feature is selected. This decomposition is exactly what isolates EFE-guided selection from the steering effect, and is discussed in Section "Multi-Action Amplification" (p. 22).

> ***Comment 8.*** *In Gemma IOI the agent selects feature steering in 94 of 100 steps after the initial stage, suggesting the high efficiency stems from repeated orientation rather than balanced exploration. The authors should discuss whether the framework is discovering circuit structure or exploiting steering-induced KL increases.*

Response: This distinction is now central to the analysis. The bounded oracle efficiency (the discovery-quality claim) is computed only from the ablation-only agent against the ablation oracle, so it cannot be inflated by steering. The steering-driven amplification is reported separately as RCK and explicitly described as exploitation of the steering action, not structure discovery (Section "Multi-Action Amplification", p. 22; Discussion "Structure Discovery versus Steering-KL Exploitation", p. 32). The action-distribution figure (Figure 10, p. 25) shows the 94/100 steering pattern and is used to explain the RCK contrast between the two models.

> ***Comment 9.*** *Appendix B gives fixed transition probabilities (ablation 50% stay / 25% down / 25% up; patching 70% stay; steering 90% stay) that are not learned from data. The authors should justify these values.*

Response: The transition probabilities are design choices encoding how strongly each action is expected to move the latent importance state, and the appendix now states this rationale: ablation can reveal or remove importance (broad, near- symmetric transitions), patching is more conservative, and steering is least likely to change the underlying importance estimate (concentrated diagonal). The full B matrices for every action are now given explicitly in the appendix ("Complete Generative Model Specification", p. 37), and a $\pm 20\%$ sensitivity sweep over the discretisation thresholds (Table 7, p. 28) together with a hyperparameter sweep over the agent's core parameters (Table 8, p. 29) shows the qualitative ordering is preserved, so the conclusions are not artefacts of these fixed values.

English language and style

The manuscript has been copy-edited throughout for concision and a consistent third-person scientific register, and overly long sentences flagged in review were split.